Lesson 8 — Responsive Design

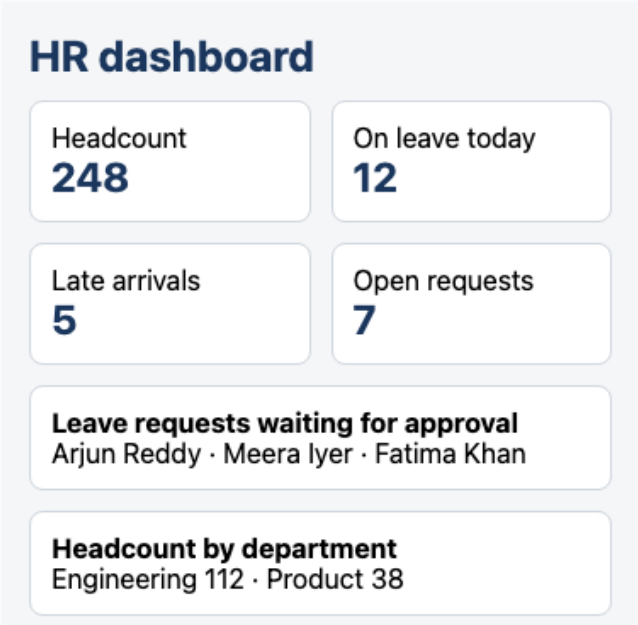
Course: HTML & CSS for Full Stack · **Session 8 of 9** · Learning path · **Previous:** Lesson 7 · **Next:** Lesson 9 — Polish, Accessibility & Final Project

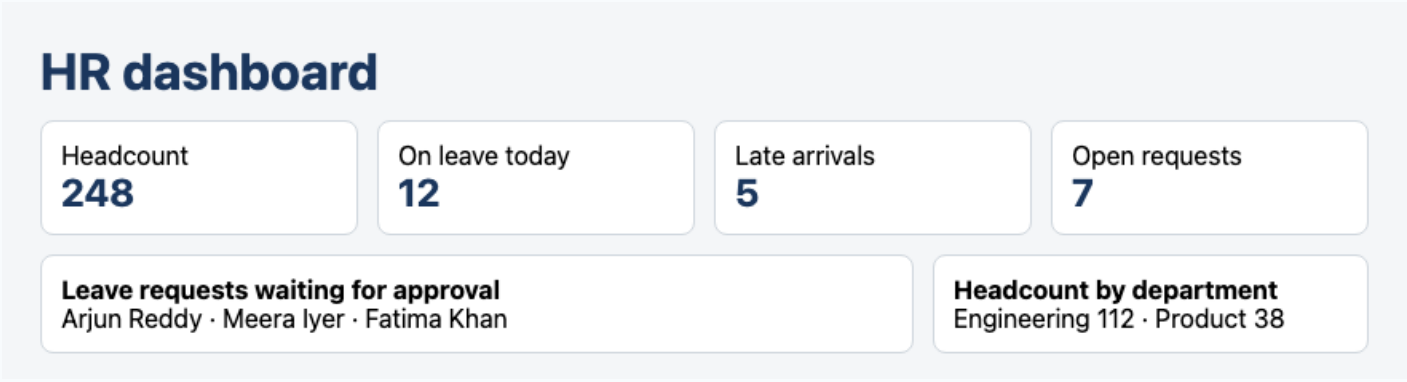
> **Module Outcome:** You can make every AcmeHR page work on a 375px phone and a wide desktop, using mobile-first media queries, fluid layouts, fluid type and responsive images.

You will build	Responsive versions of the AcmeHR header, directory, dashboard and pages
New HTML	`picture`, `source` (`media`, `srcset`)
New CSS	`@media (min-width: …)`, `min()`, `clamp()`, units `vw` and `ch`, `max-width: 100%` + `height: auto` for images
You need	Lessons 1–7

From the requirements. AcmeHR must be usable by every employee, and the BRD targets over 90% self-service adoption. Many employees will open it on a phone — to apply for leave or check attendance away from their desk.

Where we are going — the same dashboard at 375px and at 880px:





All example files: `examples/lesson08/`. To test them, make your browser window narrow, or open the page on your phone.

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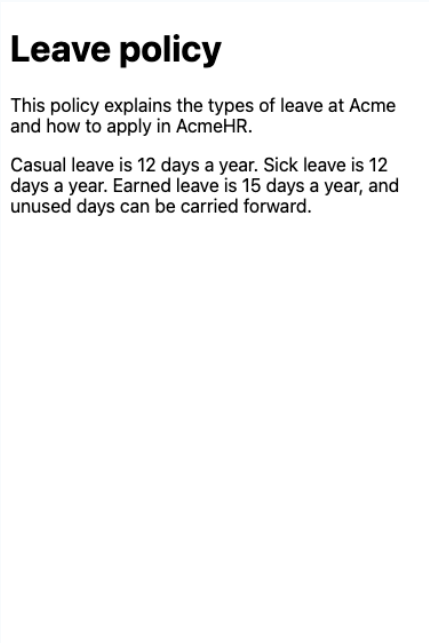
1. The viewport tag — the first line of responsive design

You have written this line since Lesson 1. Here is what it does. The same page on a 375px phone, without and with it:

Without `<meta name="viewport">`:



With `<meta name="viewport" content="width=device-width, initial-scale=1.0">`:



Without the tag, phones pretend to be a desktop screen (usually 980px wide) and shrink the result. With it, the page is laid out at the phone's real width. **No media query works properly without this tag.**

2. Fixed widths break on phones

```
.card { width: 560px; padding: 16px; border: 1px solid #d0d7de; border-radius: 8px; }
```

Output at 375px — the red dashed line is the edge of the phone screen:



The fix you already know from Lesson 3: `max-width: 560px` instead of `width: 560px`. The card is never wider than 560px, but shrinks on a small screen.

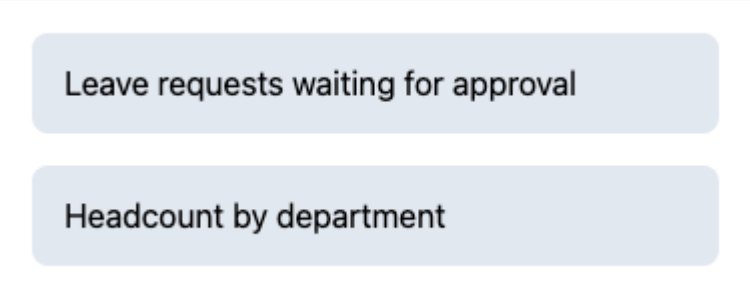
Remember: Most responsive design is not media queries. It is avoiding fixed widths: `max-width`, `%`, `fr`, `flex-wrap`, and `auto-fit`.

3. Mobile first — `@media (min-width: …)`

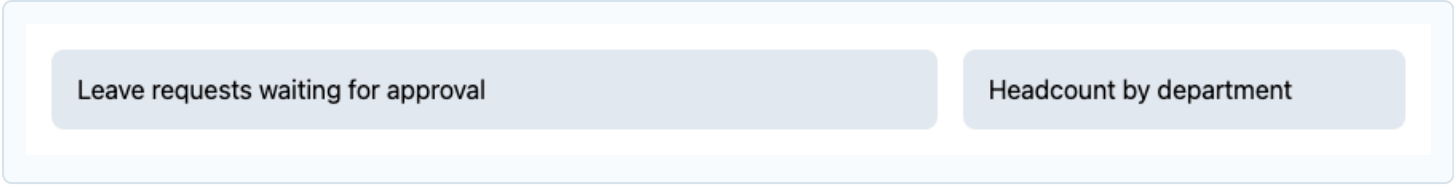
```
/* Base: phones — one column */
.layout {
  display: grid;
  gap: 16px;
}

/* From 640px wide: two columns */
@media (min-width: 40rem) {
  .layout {
    grid-template-columns: 2fr 1fr;
  }
}
```

Output at 375px:



Output at 880px:



- A **media query** applies the rules inside it only when its condition is true.
- `(min-width: 40rem)` — when the screen is at least 40rem (640px) wide.
- **Mobile first:** write the simple phone layout as the base, then **add** columns for wider screens. Phones get less CSS to apply, and you never have to undo desktop rules.
- `rem` in a media query means 16px × 40 = 640px (unless the user enlarged their default font, in which case the layout switches later — which is what they need).

The width where the layout changes is a **breakpoint**. Choose it where your content stops fitting, not by phone model. One or two breakpoints are enough for most pages.

4. The responsive header

In Lab 07 the header's links overflowed a phone screen. Mobile first fixes it:

```
/* Base: phones — logo on its own line, links wrap underneath */
header {
  padding: 8px 16px;
  background-color: #1a365d;
  color: #ffffff;
}
header p { margin: 0 0 4px; font-weight: bold; }
nav ul {
  display: flex;
  flex-wrap: wrap;
  gap: 4px 16px;
  list-style: none;
  margin: 0;
  padding: 0;
}
nav a {
  display: block;
  padding: 8px 0;
  color: #ffffff;
  text-decoration: none;
}

/* From 640px wide: logo left, links right, one line */
@media (min-width: 40rem) {
  header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 0 24px;
  }
  header p { margin: 0; }
  nav a { padding: 12px 0; }
}
```

Output at 375px:



Output at 880px:

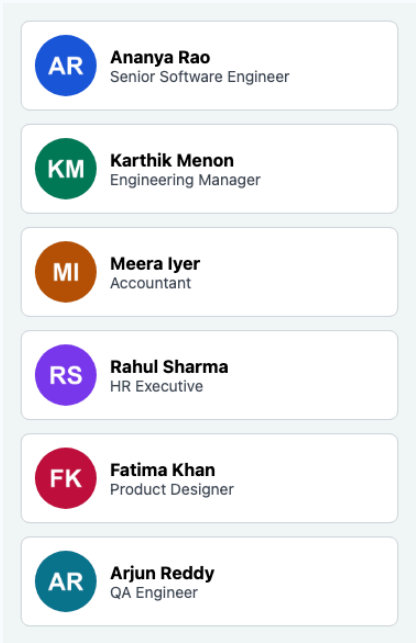


- `flex-wrap: wrap` lets the links continue on the next line instead of overflowing.
- `gap: 4px 16px` — 4px between rows, 16px between links.
- Each link keeps `8px` of padding top and bottom, so it is a comfortable target for a thumb.

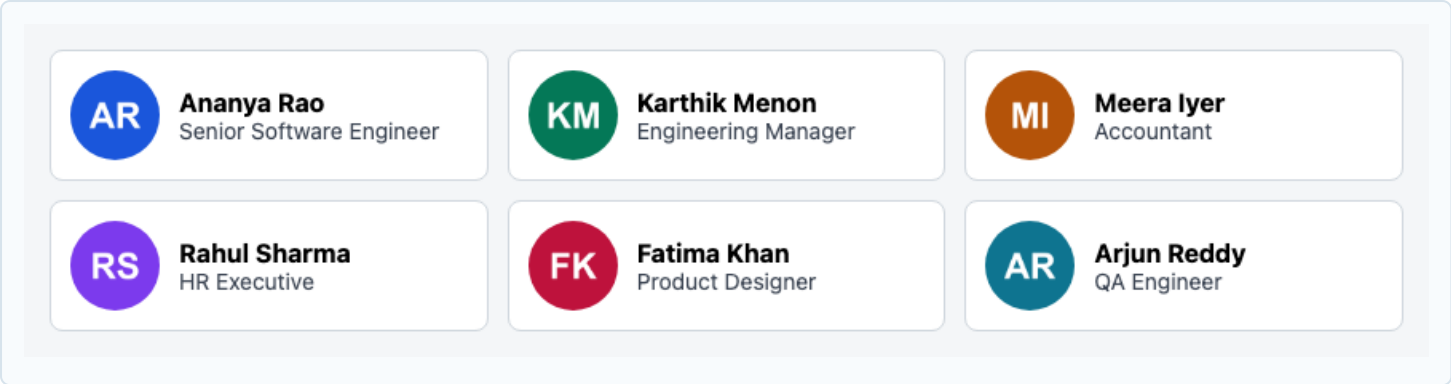
5. Layouts that adapt without a media query

```
.cards {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(min(220px, 100%), 1fr));
  gap: 12px;
}
```

Output at 375px:



Output at 880px:

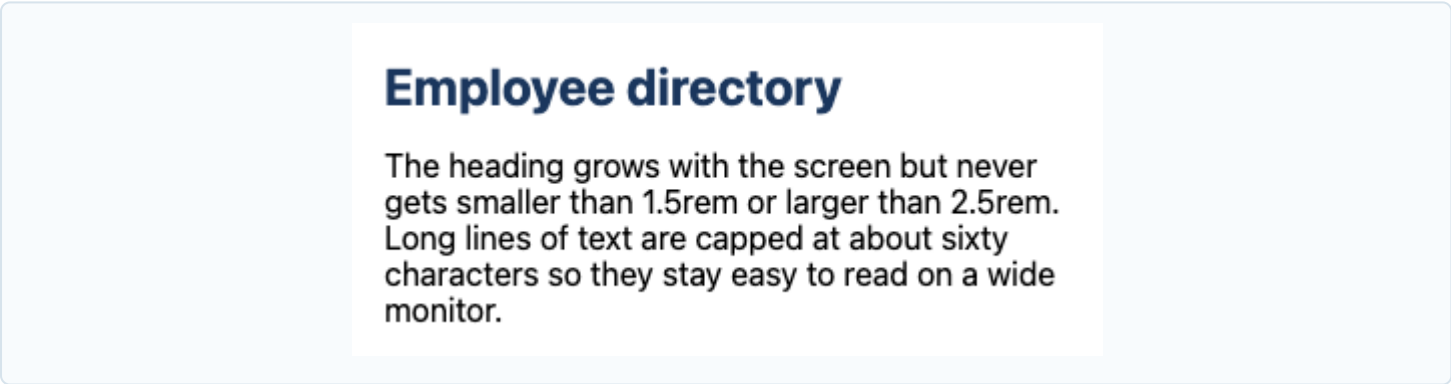


- `auto-fit` + `minmax` from Lesson 5 already adapts to the width.
- `min(220px, 100%)` picks the **smaller** of the two values. On a screen narrower than 220px plus padding, a 220px minimum would overflow; `min()` lets the column shrink to the full width instead.

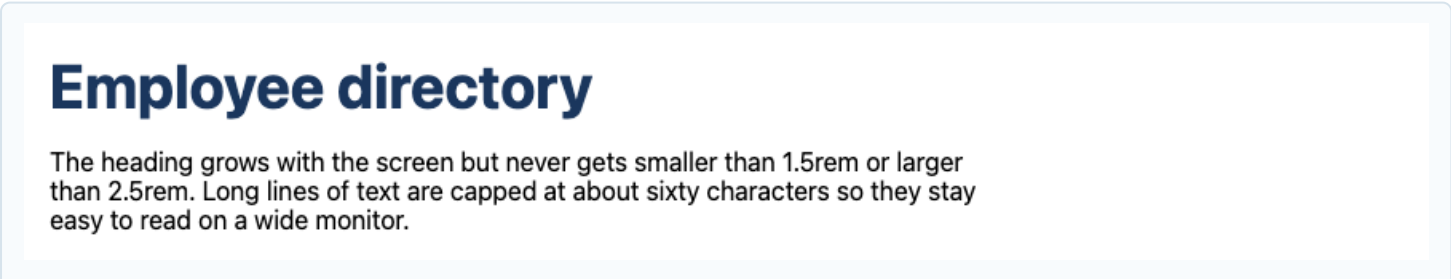
6. Fluid type — `clamp()`

```
h1 {
  font-size: clamp(1.5rem, 1rem + 2.5vw, 2.5rem);   /* at least 1.5rem, at most 2.5rem */
}
p {
  max-width: 60ch;    /* about 60 characters per line */
}
```

Output at 375px:



Output at 880px:



Value	Means
`clamp(min, preferred, max)`	Use the preferred value, but never below `min` or above `max`
`1vw`	1% of the screen's width — so `2.5vw` grows as the screen gets wider

Value	Means
`1rem + 2.5vw`	A fixed part plus a growing part. The `rem` part keeps the text zoomable.
`60ch`	About 60 characters (`ch` is the width of the "0" in the current font)

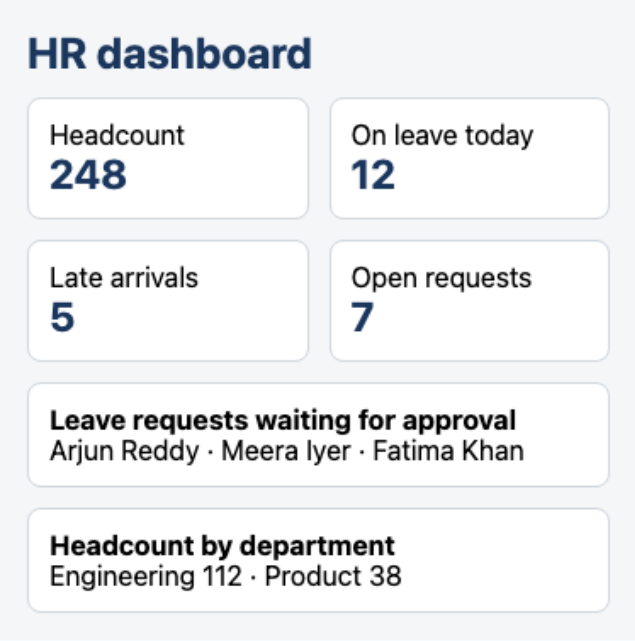
Lines of 50–75 characters are the most comfortable to read. On a wide monitor, uncapped paragraphs can be 200 characters long.

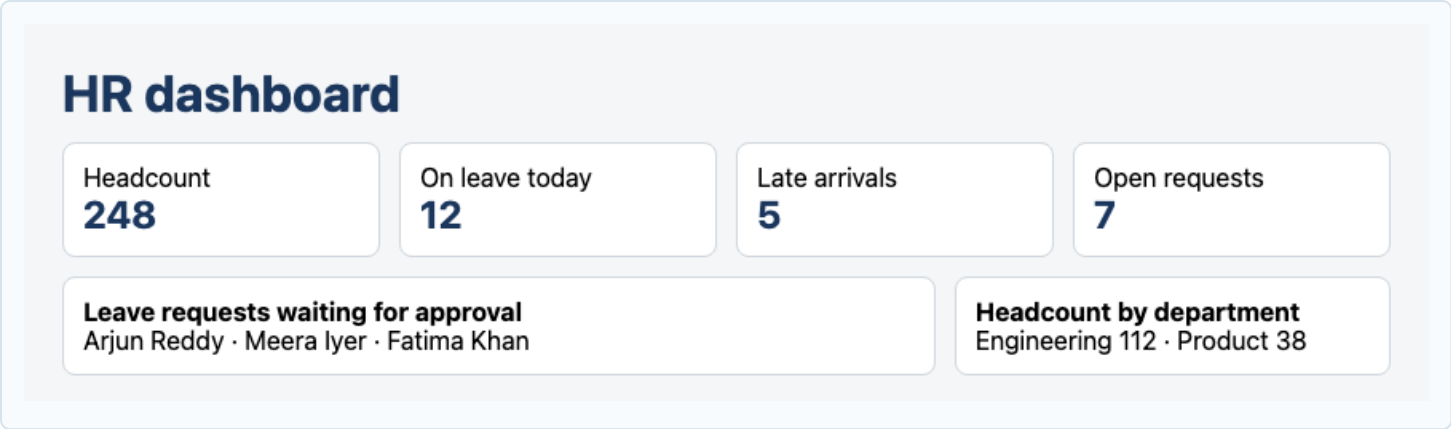
7. The dashboard — rearranging grid areas

```
.dashboard {
  display: grid;
  grid-template-areas:
    "title"
    "tiles"
    "requests"
    "departments";
  gap: 12px;
  padding: 16px;
}

@media (min-width: 40rem) {
  .dashboard {
    grid-template-columns: 2fr 1fr;
    grid-template-areas:
      "title    title"
      "tiles    tiles"
      "requests departments";
    padding: 24px;
  }
}
```

Output at 375px and at 880px:





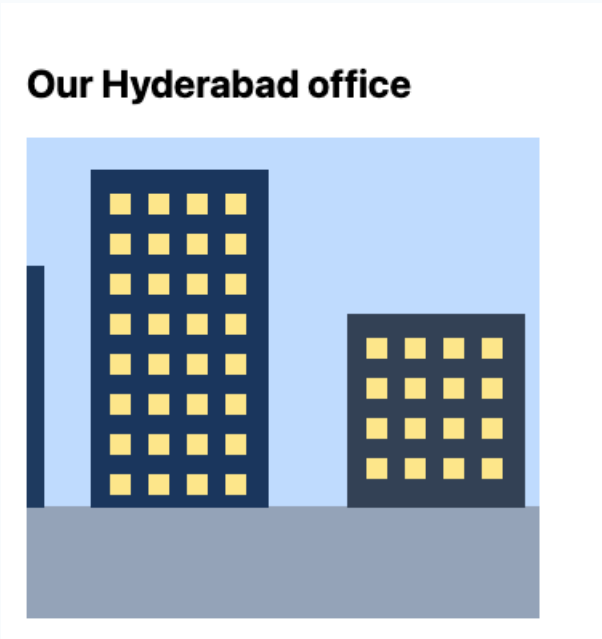
With `grid-template-areas`, the media query only has to **redraw the map**. The HTML does not change.

8. Responsive images

```
img {
  max-width: 100%;   /* never wider than its container */
  height: auto;      /* keep the shape when it shrinks */
}
```

```
<picture>
  <source media="(min-width: 40rem)" srcset="images/office-wide.png" width="640" height="320">
  <img src="images/office-tall.png" alt="Illustration of the office towers in HITEC City" width="320" he
</picture>
```

Output at 375px:



Output at 880px:

Our Hyderabad office



- `max-width: 100%; height: auto;` — the most important two lines for images. Put them in every stylesheet.
- `picture` lets you choose a **different image** for different screens. The browser uses the first `source` whose `media` matches; otherwise the `img`. Use it when a phone needs a different crop, not just a smaller size.
- The `img` inside `picture` is required — it holds the `alt` text and is the fallback.

Common Error: Forgetting the `img` inside `picture`. Nothing is shown at all.

Testing checklist for every page

Narrow your browser window slowly from wide to about 320px, and check:

1. **No sideways scrolling** of the page (a table in its own scroll box is fine).
2. **Text is readable** without zooming — at least 1rem for body text.
3. **Nothing overlaps** and nothing is cut off.
4. **Links and buttons are easy to tap** — roughly 44px tall, with space between them.
5. **Images shrink** and keep their shape.

Then open the page on your real phone. With `python3 -m http.server 8000` running, phones on the same Wi-Fi can open `http://<your-computer's-IP>:8000`.

Where used? Django and Spring send the same HTML to every device. Responsive design happens entirely in your CSS, so it works the same with any backend.

Try these

1. **(Easy)** Open the Lesson 4 directory and Lesson 5 dashboard at 375px. Write down every problem you see.
2. **(Easy)** Change the breakpoint in section 3 from `40rem` to `48rem`. At which window width does the layout now switch?
3. **(Medium)** Make the Lesson 6 leave form mobile-first: From and To stacked on phones, side by side from `40rem`.
4. **(Medium)** Give the profile page's `h1` a `clamp()` size between `1.5rem` and `2.25rem`.

5. **(Stretch)** In the dashboard, show three columns from `64rem`: requests, departments, and a new "Announcements" panel.

> **Interview:** "*What does mobile-first mean?*" — Write the base CSS for small screens, then add `min-width` media queries that enhance the layout for larger screens.

> **Interview:** "*What does the viewport meta tag do?*" — It tells mobile browsers to lay out the page at the device's real width instead of a virtual desktop width. Without it, media queries and readable text sizes do not work on phones.

> **Interview:** "*How do you make an image responsive?*" — `max-width: 100%; height: auto;`, plus `width` and `height` attributes in the HTML so space is reserved. Use `picture` with `source media` when different screens need different crops.

> **Key takeaways**
>
> - The viewport tag is required; no fixed widths — use `max-width`, `%`, `fr`.
> - Mobile first: phone layout as the base, `@media (min-width: …)` to add.
> - `flex-wrap`, `auto-fit` + `minmax(min(…, 100%), 1fr)` adapt without media queries.
> - `clamp()` for fluid headings; `max-width: 60ch` for readable lines.
> - `img { max-width: 100%; height: auto; }` everywhere; `picture` for different crops.
> - Test from wide down to 320px, then on a real phone.

Lab

👉 Lab 08 — Make AcmeHR responsive

Next: Lesson 9 — Polish, Accessibility & Final Project